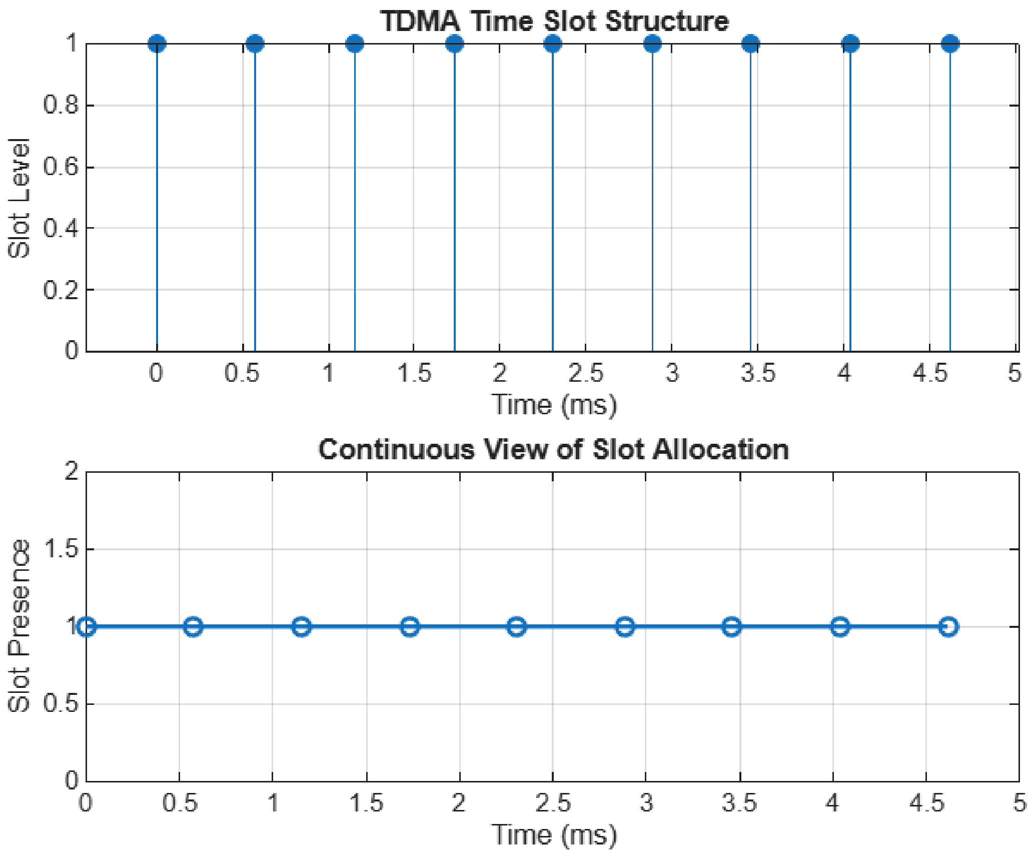
Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

OBJECTIVE 1

```
clc;
clear;
close all;
FrameTime = 4.615;
Slots = 4;        % Changed from 8 to 4
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

OUTPUT



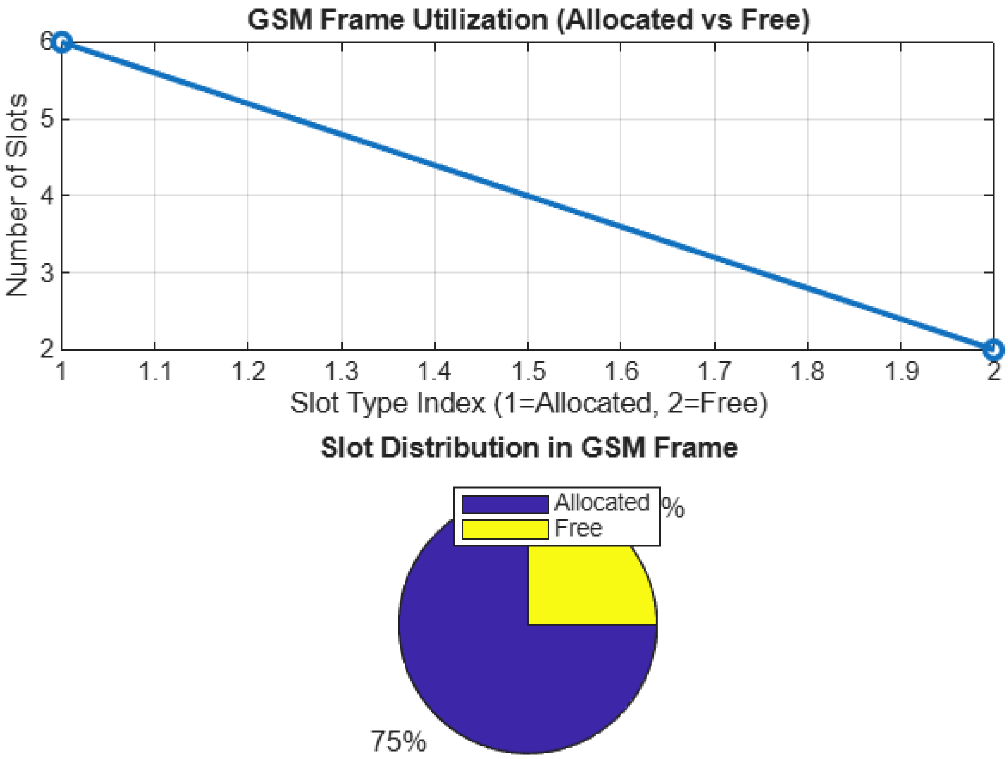
OBJECTIVE 2

```
clc;
clear;
close all;
TotalSlots = 4;
AllocatedSlots = 3;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on
% Graph 2: Pie Chart
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```

OUTPUT

```
Frame Utilization (%)
    75
```


GSM Frame Utilization (Allocated vs Free)
Number of Slots
6
5
4
3
2
1
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
2
Slot Type Index (1=Allocated, 2=Free)
Slot Distribution in GSM Frame
Allocated
Free
%
75%